Industrial Instrumentation and Process Control Technician

Full-time Diploma

Courses

The Industrial Instrumentation and Process Control Technician program contains a blend of classroom lectures and practical laboratory sessions using the latest technology. Troubleshooting exercises guide you through real-world examples and teach you how to solve problems. Students learn through hands-on work experience. Fifty percent of their learning time will be spent training in BCIT's real-world processing plants that are central to this program.

Class hours

Depending on the intakes available, class hours will be 0700–1345 or 1015–1700, Monday to Friday

Program matrix

	Term 1: (15 weeks)	Credits
INST 1000	**DC Electricity** course outline	6.0
INST 1001	**Symbols, Technical Writing & Computer Aided Design** course outline	4.0
INST 1103	**Instrumentation Fundamentals** course outline	6.0
INST 2000	**AC Electricity and Transistors** course outline	7.0
INST 2001	**Power Supplies & Operational Amplifiers** course outline	7.0

	Term 2: (15 weeks)	Credits
INST 2112	**Pressure Instruments and Measurement** course outline	8.0
INST 2116	**Flow Instruments and Measurements** course outline	6.0
INST 2121	**Level, Density Instruments and Measurements** course outline	6.0
INST 2123	**Control Valves and Positioners** course outline	8.0
INST 2125	**Hydraulic Systems** course outline	2.0

	Term 3: (15 weeks)	Credits
INST 2107	**Temperature Instruments and Measurement** course outline	6.0
INST 3112	**Programmable Logic Controllers and Human Machine Interface (HMI)** course outline	8.0
INST 3117	**Analog and Microprocessor-Based Controllers** course outline	8.0
INST 3121	**Process Control and Control Systems** course outline	8.0

	Term 4: (15 weeks)	Credits
INST 3106	**Analytical Instruments and Control** course outline	6.0
INST 3110	**Field Effect Transistors (FETs) & Thyristors, Digital Logic** course outline	6.0
INST 3124	**Distributed Control Systems** course outline	6.0
INST 3128	**Boiler Control and Foundation Fieldbus** course outline	6.0
INST 3130	**Viscosity, Consistency, Pulp Blend & Batch Control** course outline	6.0

	Total Credits:	120.0

Transfer credit

Do you have credits from another BC/Yukon post-secondary school? Do you want to know if they transfer to courses here at BCIT? Check out BCIT's Transfer Equivalency Database to find out.

Programs and courses are subject to change without notice.

TERM I

INST 1000 - DC Electricity

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	September 06, 2016
End Date:	December 16, 2016
Total Hours:	90
Total Weeks:	3
Hours/Weeks:	30
Delivery Type:	Lecture/Lab
CRN:	43103

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127
Office Hours:

Wednesdays 11:30 to 1:45

Course description

Students will learn basic electrical theory and applied mathematics for instrumentation. Topics will include series/parallel circuits, batteries, potentiometers and rheostats. Mathematical topics will include logarithms, exponents, algebra, trigonometry, rectangular and polar coordinates. Problems will be focused on various aspects of instrumentation.

Course learning outcomes / competencies

Upon successful completion of this course, the student will be able to:

Safety

- Demonstrate safe working practices in the lab

DC Electricity and Mathematics

- Explain basic concepts of electrical theory
- Apply laws of DC electricity in circuits
- Evaluate battery circuits
- Design potentiometer and rheostat circuits
- Build DC circuits and perform necessary troubleshooting to ensure proper circuit function
- Solve algebra and geometry problems in the context of instrumentation
- Solve instrumentation problems involving periodic, logarithmic, exponential, and trigonometric functions

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days without justification, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100 %	

Attendance requirements

Regular attendance in lectures and labs is seen as integral to student success, and will be monitored. Students are expected to be punctual and attend class for the full duration. Late arrivals or early departures may be considered as absences. Absence from more than 10% of the scheduled course hours will result in the issuance of a warning. If additional scheduled sessions are missed, withdrawal or course failure may be recommended.

Learning resources

Required materials will be distributed by the instructor.

Safety equipment or protective clothing

Safety boots

Safety glasses

Course schedule and assignments

Monday to Friday

Lecture 0700 to 0830; 0845 to 1015

Lab 1045 to 1345

When appropriate, all-day lectures or all-day labs will be scheduled

INST 2000 - AC Electricity and Transistors

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	7
Minimum Passing Grade:	70%
Start Date:	September 06, 2016
End Date:	December 16, 2016
Total Hours:	105
Total Weeks:	4
Hours/Weeks:	26.25
Delivery Type:	Lecture/Lab
CRN:	43104

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127
Office Hours:

Wednesdays 11:30 to 1:45

Course description

Students will be exposed to AC transformer theory including peak, average and RMS voltage and current. Labs will include transformers, capacitors and inductors. Transistor theory and labs will include RC coupled amplifiers, common emitter, common collector and common base problems.

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

- Safety
- Demonstrate safe working practices in the lab

AC Electricity

- Explain basic concepts of AC electricity.
- Describe various types of transformers.
- Explain the purpose of capacitors and inductors in AC circuits.
- Size electrical components for various circuits.
- Build and test AC circuits that include transformers, resistors, capacitors, and inductors.

Transistors

- Explain the key concepts of Bipolar Junction Transistor (BJT) theory.
- Explain BJT functionality.
- Solve transistor circuits that include bipolar junction transistors, capacitors and resistors.
- Assemble various transistor circuits.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days without justification, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100 %	

Attendance requirements

Regular attendance in lectures and labs is seen as integral to student success, and will be monitored. Students are expected to be punctual and attend class for the full duration. Late arrivals or early departures may be considered as absences. Absence from more than 10% of the scheduled course hours

will result in the issuance of a warning. If additional scheduled sessions are missed, withdrawal or course failure may be recommended.

Learning resources

Required materials will be distributed by the instructor.

Safety equipment or protective clothing

Safety boots

Safety glasses

Course schedule and assignments

Monday to Friday

Lecture 0700 to 0830; 0845 to 1015

Lab 1045 to 1345

When appropriate, all-day lectures or all-day labs will be scheduled

INST 2001 Power Supplies and Operational Amplifiers

Students will explain and build various DC power supplies and Operational Amplifiers (OpAmps) circuits. Topics will include oscilloscope operation and applications. Troubleshooting will include OpAmp circuits and power supply circuits.

TERM II

INST 2112 Pressure Instruments and Measurement

Course learning outcomes / competencies

- Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Pressure Measurement

- Convert pressure measurements among three standards: SI, MKS and BGS
- Define pressure types, such as atmospheric, absolute, gauge and vacuum.
- Identify zero, span & angularity errors when adjusting calibration to correct errors in gauges.
- Calibrate gauges using primary standards such as manometers and dead weight testers which are primary standards.
- Identify pressure gauges by type, class, application and gauge accessories, such as pulsation dampers, chemical seals and pigtail siphons.
- Calculate and correct for head effects on pressure and measurement.
- Identify receiver elements.
- Describe pressure switch contacts, such as mercury and micro switches.
- Calibrate pressure switches using appropriate test standards.

Pneumatic and Electronic Instruments

- Use a dead weight tester or a manometer to calibrate different types of gauges such as draft gauge, standard pressure gauge and high pressure gauge to manufacturer’s specified accuracy.
- Use the appropriate calibration test equipment to calibrate recorders including servo-mechanisms and other link and lever devices to manufacturer’s specified accuracy.
- Use the appropriate calibration test equipment to calibrate pressure pneumatic, electronic and smart electronic pressure transmitters using appropriate test equipment to manufacturer’s specified accuracy.
- Perform the set-up of a pneumatic relay.

- Explain the function of pressure measuring elements, such as bellows, diaphragm and helical.
- Describe force balance transmitters such as moment of force and true force.
- Describe motion balance transmitters such as angle motion and linear motion balance.
- Calibrate transducers (pneumatic to current and current to pneumatic).

INST 2116 Flow Instruments and Measurements

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Density Measurement

- Define density and units of density, such as API, Baumé, BRIX and TWADDLE.
- Describe hydrometers and boiling point rise used for density measurement.
- Describe hydrostatic head density, various means of achieving hydrostatic density measurement and displacement density measurement.
- Explain nuclear density meter operation and the Atomic Energy of Canada (AEC) safety regulation related to nuclear instrumentation.
- Perform the yearly test required as per AEC guidelines on nuclear instruments using Geiger counter and swab test to ensure the integrity of nuclear instruments.
- Calibrate nuclear density instrument transmitter.
- Calibrate a refractometer.
- Describe boiling point rise to density measurement and linearization of boiling point rise to density.
- Perform an empirical test and calibration of boiling point rise systems on running multiple effect evaporators.

Level Measurement

- Describe level switch contacts, such as mercury, micro switches and gauge glasses.
- Define hydrostatic head level measurement and 'elevation' and 'suppression' on head level measurements.

- Describe open and closed tank head level measurement, such as condensing and non-condensing head level measurement.
- Calibrate head level measurement transmitters.
- Calculate bubble pipe level measurements.
- Calculate displacer sizes for displacement level transmitters.
- Describe nuclear, capacitance, ultrasonic, pulse radar, load cell and boiler drum level measurement.
- Calibrate ultrasonic level measurement.
- Calculate zero, span, elevation and suppression parameters as they apply to head level measurement of various processes such as distillation column, multiple effect evaporators, industrial boilers etc.
- Install a level transmitter on a tank.

INST 2121 Level, Density Instruments and Measurements

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Density Measurement

- Define density and units of density, such as API, Baumé, BRIX and TWADDLE.
- Describe hydrometers and boiling point rise used for density measurement.
- Describe hydrostatic head density, various means of achieving hydrostatic density measurement and displacement density measurement.
- Explain nuclear density meter operation and the Atomic Energy of Canada (AEC) safety regulation related to nuclear instrumentation.
- Perform the yearly test required as per AEC guidelines on nuclear instruments using Geiger counter and swab test to ensure the integrity of nuclear instruments.
- Calibrate nuclear density instrument transmitter.
- Calibrate a refractometer.

- Describe boiling point rise to density measurement and linearization of boiling point rise to density.
- Perform an empirical test and calibration of boiling point rise systems on running multiple effect evaporators.

Level Measurement

- Describe level switch contacts, such as mercury, micro switches and gauge glasses.
- Define hydrostatic head level measurement and 'elevation' and 'suppression' on head level measurements.
- Describe open and closed tank head level measurement, such as condensing and non-condensing head level measurement.
- Calibrate head level measurement transmitters.
- Calculate bubble pipe level measurements.
- Calculate displacer sizes for displacement level transmitters.
- Describe nuclear, capacitance, ultrasonic, pulse radar, load cell and boiler drum level measurement.
- Calibrate ultrasonic level measurement.
- Calculate zero, span, elevation and suppression parameters as they apply to head level measurement of various processes such as distillation column, multiple effect evaporators, industrial boilers etc.
- Install a level transmitter on a tank.

INST 2125 Hydraulic Systems

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Hydraulics

- Describe the function of different elements of hydraulic systems.
- Explain hydraulic specifications and safety hazards.
- Assemble various hydraulic circuits.

- Test the previously assembled hydraulic circuits.
- Troubleshoot any issues resulting from the testing of hydraulic circuits.

INST 2123 Control Valves and Positioners

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Valve Positioners

- Identify control valves by actuator and body type.
- Describe process applications for various valves.
- Service (disassemble/inspect/reassemble/test) control valves.
- Stroke control valves to ensure accurate process operation.
- Explain the application of positioners for sliding stem valves and rotary action valves.
- Describe installation requirements and considerations for positioners, such as air supply, signal by-pass switches, reverse-acting and bench set.
- Calibrate and stroke smart positioners, such as Fisher DVC, Hartman Braun.
- Explain the basic architecture of a Foundation Fieldbus system as it applies to valve positioners.
- Configure Foundation Fieldbus DVC positioners applied to an operating control system.

Valve Sizing

- Use a pump curve to determine net positive suction head (NPSH) and discharge head.
- Describe valve sizing for liquids including viscosity effects, steam and gas.
- Calculate appropriate valve sizes for various process applications.
- Describe the maximum d/p allowable parameter applied to valve sizing.
- Select appropriate valve trims for cavitation and noise reduction applications.

TERM III

INST 2107 - Temperature Instruments and Measurement

School:	School of Energy
Program:	IIPCT
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	September 05, 2017
End Date:	December 15, 2017
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	44412

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127

Course description

This course examines temperature units and conversion. Topics covered also include principles of temperature measurement such as: thermometers and bi-metallic, filled thermal systems, temperature switches, RTD's (includes thermistors), thermocouples and non-contract temperature measurements. Students will be introduced to the various temperature instruments such as: pneumatic transmitters, electronic transmitters, smart transmitters (digital), temperature recorders and current to current transducers. Prerequisite: Successful completion of term 2 courses

Course goals

Upon successful completion, the student will be able to:

Safety

1. Demonstrate safe working practices in the lab.

Temperature Measurement

1. Define temperature and heat transfer.
2. Identify thermometer types.

3. Identify types of filled thermal systems and bi-metallic temperature elements.
4. Describe types and classes of filled thermal systems, location of temperature measuring elements and switch contacts, such as mercury and micro-switches.
5. Calibrate temperature switches using appropriate test standards.
6. Explain resistance temperature detectors, the different types of RTDs, such as platinum and nickel and 2-wire, 3-wire and 4-wire RTD measurement.
7. Calibrate RTD temperature instruments using a calibrator and resistance tables.
8. Explain thermocouple theory.
9. Identify types of thermocouples used in temperature measurement.
10. Calculate the millivoltage for temperature readings and calibration for common types of thermocouples, such as J, K, T and E.
11. Describe non-contact temperature measurement system, such as optical pyrometers and thermistors.

Temperature Instruments

1. Describe various types of transmitters, such as pneumatic, electrical and smart.
2. Describe temperature records.
3. Describe current to current transducers.

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety:

- Demonstrate safe working practices in the lab.

Temperature Measurement:

- Define temperature and heat transfer.
- Identify thermometer types.
- Identify types of filled thermal systems and bi-metallic temperature elements.
- Describe types and classes of filled thermal systems, location of temperature measuring elements and switch contacts, such as mercury and micro-switches.
- Calibrate temperature switches using appropriate test standards.

- Explain resistance temperature detectors, the different types of RTDs, such as platinum and nickel and 2-wire, 3-wire and 4-wire RTD measurement.
- Calibrate RTD temperature instruments using a calibrator and resistance tables.
- Explain thermocouple theory.
- Identify types of thermocouples used in temperature measurement.
- Calculate the millivoltage for temperature readings and calibration for common types of thermocouples, such as J, K, T and E.
- Describe non-contact temperature measurement system, such as optical pyrometers and thermistors.

Temperature Instruments:

- Describe various types of transmitters, such as pneumatic, electrical and smart.
- Describe temperature records.
- Describe current to current transducers.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days without justification, they may be dismissed from the program Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

INST 3112 - Programmable Logic Controllers and Human Machine Interface (HMI)

School: School of Energy

Program:	IIPCT
Course Credits:	8
Minimum Passing Grade:	70%
Start Date:	September 05, 2017
End Date:	December 15, 2017
Total Hours:	120
Total Weeks:	15
Hours/Weeks:	8
Delivery Type:	Lecture/Lab
CRN:	44413

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127

Course description

This course will prepare students to be able to design and interpret conventional relay logic. They will also learn how to configure, program and troubleshoot Programmable Logic Controllers (PLCs) for motor control. Students will also learn to configure, program and interface with PLC's using Human Machine Interface (HMI) software. Prrequisite: Successful completion of term 2 courses

Course goals

Upon successful completion, the student will be able to:

Safety

1. Demonstrate safe working practices in the lab.

Relays & Ladder Logic

1. Explain electrical relay types including on-delay and off-delay timers with ratings for both AC & DC
2. Build circuits using on-delay and off-delay timers and relays
3. Troubleshoot circuits that include on-delay and off-delay timers and relays

Communication Basics

1. Explain various communication network topographies, standards and protocols such as RS232, RS485, Modbus and Ethernet to communicate with PLCs and HMIs
2. Explain the Open Systems Integration (OSI) model to modern communication systems
3. Configure a typical SCADA system with I/O

4. Explain PLC system grounding requirements, independent from electrical grounding requirements

Programming of PLCs using Boolean Instructions

1. Explain the operation of small ('shoebox') PLCs with Boolean programming instructions
2. Program PLCs using Boolean instructions.
3. Troubleshoot the PLC software and field circuits

Programmable Logic Control using Ladder Logic

1. Describe the architecture of a PLC such as a Modicon Quantum including various types of I/O modules
2. Program various motor control circuits using ladder logic
3. Wire a PLC to field equipment demonstrating safe wiring practices
4. Explain the difference between 2-wire and 3-wire motor control circuits

Human Machine Interface (HMI)

1. Create an application in an HMI environment
2. Create an I/O database with animation effects to interface with the PLC database

Program a single loop level control project complete with trending, pump, mixer, interlocks and alarm

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Relays & Ladder Logic

- Explain electrical relay types including on-delay and off-delay timers with ratings for both AC & DC
- Build circuits using on-delay and off-delay timers and relays
- Troubleshoot circuits that include on-delay and off-delay timers and relays

Communication Basics

- Explain various communication network topographies, standards and protocols such as RS232, RS485, Modbus and Ethernet to communicate with PLCs and HMIs.

- Explain the Open Systems Integration (OSI) model to modern communication systems.
- Configure a typical SCADA system with I/O.
- Explain PLC system grounding requirements, independent from electrical grounding requirements.

Programming of PLCs using Boolean Instructions

- Explain the operation of small ('shoebox') PLCs with Boolean programming instructions.
- Program PLCs using Boolean instructions.
- Troubleshoot the PLC software and field circuits.

Programmable Logic Control using Ladder Logic

- Describe the architecture of a PLC such as a Modicon Quantum including various types of I/O modules.
- Program various motor control circuits using ladder logic.
- Wire a PLC to field equipment demonstrating safe wiring practices.
- Explain the difference between 2-wire and 3-wire motor control circuits.

Human Machine Interface (HMI)

- Create an application in an HMI environment.
- Create an I/O database with animation effects to interface with the PLC database.
- Program a single loop level control project complete with trending, pump, mixer, interlocks and alarms.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program

- Participation is based on initiative, attitude, workmanship and effective communication in classes and labs

Quizzes	5
Labs	25
Participation	10
Final Exam	30

INST 3117 - Analog and Microprocessor-Based Controllers

School:	School of Energy
Program:	IIPCT
Course Credits:	8
Minimum Passing Grade:	70%
Start Date:	September 05, 2017
End Date:	December 15, 2017
Total Hours:	120
Total Weeks:	15
Hours/Weeks:	8
Delivery Type:	Lecture/Lab
CRN:	44414

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127

Course description

Students will learn the fundamentals of signal conditioning and scaling, the fundamentals of control modes and controller modes and the alignment and maintenance of pneumatic controllers. As part of course work, students will program microprocessor controllers and wire them for a given application.
Prerequisite: Successful completion of term 2 courses

Course goals

Upon successful completion, the student will be able to:

Safety

1. Demonstrate safe working practices in the lab.

Signal Conditioning and Scaling

1. Explain the application of high and low select relays, limit relays or function blocks.
2. Explain the application of signal reversing relays, amplifying and/or reducing relays, signal summing relays or function blocks.
3. Calculate totals using the scaling of signals so that they can be meaningfully computed.

Control Modes

1. Explain On/Off control action, single speed floating control (integral) and differential gap control applied to industrial processes.

Controller Modes

1. Explain the operation of proportional, integral, derivative (PID) control actions applied to industrial processes.
2. Calculate the proportional response based on various deviations, actions and various proportional bands or gain settings.
3. Calculate the P+I response given various deviations, actions; PBs and reset settings.
4. Explain derivative action including both rate applied to PV and rate applied to error responses to process variations.
5. Explain interacting and non-interacting PID algorithms.
6. Determine the correct controller actions and the appropriate modes of control for various process installations.

Controller Alignment and Maintenance

1. Explain the automatic and manual controller operation including bumpless, balanceless automatic/manual transfer for pneumatic, microprocessor stand-alone controllers and various DCS/PLC systems.
2. Explain the operation of remote and local set point and the application of set point tracking and output tracking.
3. Perform and test controller alignment on pneumatic field and panel mounted controllers.

Electronic Analog & Microprocessor Based Controllers

1. Explain the principles of microprocessor-based controllers using function block programming and/or configuration menus.
2. Configure microprocessor-based controllers for typical process applications such as single loop control, cascade control and feedforward control.
3. Create a P and ID drawing of a given process application.

4. From a P and ID drawing, build a complete control loop to include sensing element, transmitter, controller, recorder, transducer and final control element.

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Signal Conditioning and Scaling

- Explain the application of high and low select relays, limit relays or function blocks.
- Explain the application of signal reversing relays, amplifying and/or reducing relays, signal summing relays or function blocks.
- Calculate totals using the scaling of signals so that they can be meaningfully computed.

Control Modes

- Explain On/Off control action, single speed floating control (integral) and differential gap control applied to industrial processes.

Controller Modes

- Explain the operation of proportional, integral, derivative (PID) control actions applied to industrial processes.
- Calculate the proportional response based on various deviations, actions and various proportional bands or gain settings.
- Calculate the P+I response given various deviations, actions; PBs and reset settings.
- Explain derivative action including both rate applied to PV and rate applied to error responses to process variations.
- Explain interacting and non-interacting PID algorithms.
- Determine the correct controller actions and the appropriate modes of control for various process installations.

Controller Alignment and Maintenance

- Explain the automatic and manual controller operation including bumpless, balanceless automatic/manual transfer for pneumatic, microprocessor stand-alone controllers and various DCS/PLC systems.

- Explain the operation of remote and local set point and the application of set point tracking and output tracking.
- Perform and test controller alignment on pneumatic field and panel mounted controllers.

Electronic Analog & Microprocessor Based Controllers

- Explain the principles of microprocessor-based controllers using function block programming and/or configuration menus.
- Configure microprocessor-based controllers for typical process applications such as single loop control, cascade control and feedforward control.
- Create a P and ID drawing of a given process application.
- From a P and ID drawing, build a complete control loop to include sensing element, transmitter, controller, recorder, transducer and final control element.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70%
Quizzes	5	• Students will work on individual projects in labs
Labs	25	• There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately
Participation	10	• Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program
Final Exam	30	• Participation is based on initiative, attitude, workmanship and effective communication in classes and labs

INST 3121 - Process Control and Control Systems

School:	School of Energy
Program:	IIPCT
Course Credits:	8
Minimum Passing Grade:	70%
Start Date:	September 05, 2017
End Date:	December 15, 2017
Total Hours:	120

Total Weeks:	15
Hours/Weeks:	8
Delivery Type:	Lecture/Lab
CRN:	44415

Instructor Details

Name: John Powszedny
Email: jpowszedny@bcit.ca
Location: Se-8 Rm 127

Course description

This course looks at a variety of process control and control systems. Analog control concepts such as on/off, two mode control, three mode control. Pneumatic controllers, electronic controllers and manual/auto transfer are investigated. The following control strategies are also covered: duplex control, auto selector, gap action, relation, cascade, ratio and feed forward. Loop tuning techniques and hands-on experience with live processes are included. Variable Speed Controllers (VSD's or VFD's) are also included as they are an alternative Final Control Element (FCE) and are commonly used to improve efficient use of energy in modern control systems. Prerequisite: Successful completion of term 2 courses

Course goals

Upon successful completion, the student will be able to:

Safety

1. Demonstrate safe working practices in the lab.

Control Systems

1. Design a variety of control systems including cascade control with appropriate variable, gap action, duplex, auto-selection and feedforward index systems.

Process Control

1. Explain the process thermodynamics and heat transfer.
2. Use steam tables to determine process thermodynamics and energy available for heat transfer.
3. Explain material and energy balance of an operating double effect evaporator process using ethylene glycol.
4. Explain natural gas production and processing using glycol dehydration (TEG and DEG).

Loop Tuning

1. Explain various degrees of process self-regulation and the interaction of controller modes.
2. Explain Ziegler Nichols, Lambda tuning techniques and tuning from manual output changes.

3. Calculate initial PID settings to tune loops on various live processes including evaporation, material blending (pressure/flow) and chemical (cascade steam heat exchangers).
4. Explain process coupling problems and the techniques for decoupling sympathetic control loops.

Variable Speed Controls (VFDs)

1. Explain motor/pump speed control as an alternative to a control valve.
2. Explain variety of variable speed devices.

- Variable Pitch
- Eddy Current Couplings
- Variable Frequency Drives

1. Explain basics of a three phase induction motor including ratings, service applications, inrush current and its ramifications.
2. Explain methods for AC motor speed control through pulse width modulation (PWM).
3. Configure VFD for motor speed and torque control on various loads.

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Control Systems

- Design a variety of control systems including cascade control with appropriate variable, gap action, duplex, auto-selection and feedforward index systems.

Process Control

- Explain the process thermodynamics and heat transfer.
- Use steam tables to determine process thermodynamics and energy available for heat transfer.
- Explain material and energy balance of an operating double effect evaporator process using ethylene glycol.
- Explain natural gas production and processing using glycol dehydration (TEG and DEG).

Loop Tuning

- Explain various degrees of process self-regulation and the interaction of controller modes.
- Explain Ziegler Nichols, Lambda tuning techniques and tuning from manual output changes.
- Calculate initial PID settings to tune loops on various live processes including evaporation, material blending (pressure/flow) and chemical (cascade steam heat exchangers).
- Explain process coupling problems and the techniques for decoupling sympathetic control loops.

Variable Speed Controls (VFDs)

- Explain motor/pump speed control as an alternative to a control valve.
- Explain variety of variable speed devices.
    - Variable Pitch
    - Eddy Current Couplings
    - Variable Frequency Drives
- Explain basics of a three phase induction motor including ratings, service applications, inrush current and its ramifications.
- Explain methods for AC motor speed control through pulse width modulation (PWM).
- Configure VFD for motor speed and torque control on various loads.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on assigned labs in teams • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

TERM IV

INST 3106 - Analytical Instruments and Control

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	January 08, 2018
End Date:	April 20, 2018
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	85250

Instructor Details

Name: Ron Vandergaag
Email: Ron_Vandergaag@bcit.ca

Course description

Students will be trained in the fundamentals and calibration of transmitters used for analytical measurement such as pH, conductivity, ORP and dissolved oxygen. Students will gain experience tuning DCS control systems using advanced control strategies to operate a chemical process to control variables such as pH and electrolytic conductivity. Prerequisite: Successful completion of term 3 courses

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Analytical Measurement and Control

- Explain the complexity of the control of the pH measurement in industrial processes.
- Explain various pH measurement components such as glass, reference, temperature compensation electrodes and transmitters.
- Calibrate various pH transmitters complete with their associated electrodes.
- Explain the difference between absolute and relative electrolytic conductivity measurement.

- Calibrate various conductivity transmitters and their associated electrodes to include contacting and non-contacting systems.
- Explain oxidation reduction potential (ORP) measurement and applications such as cyanide destruction, hexavalent chromium destruction, sewage treatment and pulp bleaching.
- Calibrate dissolved oxygen and turbidity meters including the transmitters and the sensors.
- Develop a graph from the titration of a chemical reaction to determine the breakpoints and tuning parameters for a non-linear adaptive gain control system.
- Using the Foxboro I/A DCS system, operate and tune a chemical control process to include loops such as temperature, level, flow, pH (non-linear adaptive gain and feedforward indexed loops etc.), conductivity and dissolved oxygen.

Evaluation criteria

Criteria **% Comments**

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on assigned labs in teams • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

INST 3110 - Field Effect Transistors (FETs) & Thyristors, Digital Logic

School: School of Energy
Program: Industrial Instrumentation Process and Control Technician
Course Credits: 6
Minimum Passing Grade: 70%

Start Date:	January 08, 2018
End Date:	April 20, 2018
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	85251

Instructor Details

Name: Ron Vandergaag
Email: Ron_Vandergaag@bcit.ca
Location: SE-8 Rm 127

Course description

In this course students will learn the theory and application of different types of flip flop circuits. Basic Thyristor theory including Field Effect Transistors (FETs) and Silicon Control Rectifiers (SCRs) will be covered. Boolean logic including Karnaugh maps and Boolean PLCs will be covered in detail in the lab. Prerequisite: Successful completion of term 3 courses

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Field Effect Transistors (FETs) & Thyristors

- Explain the functionality of Light Emitting Diodes (LEDs), Photodiodes & Optical Isolators.
- Describe the functionality of various Thyristors (Silicon Controlled Rectifiers (SCRs), Triacs and Silicon Controlled Switches (SCS), Field Effect Transistors (FETs) and MOSFETs).
- Design LED and SCR circuits.
- Build LED and SCR circuits.

Digital Logic

- Explain different number systems (binary, octal, hexadecimal).
- Design various types of flip flop circuits.
- Create timing diagrams using appropriate symbols.
- Explain common Boolean identities and solve Boolean equations.
- Describe various types of logic gates (e.g. AND, OR, NAND, NOR, NOT, XOR).

- Solve Karnaugh Maps.
- Design flip flop and Boolean logic circuits.
- Build flip flop and Boolean logic circuits.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

INST 3124 - Distributed Control Systems

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	January 08, 2018
End Date:	April 20, 2018
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	85252

Instructor Details

Name: Ron Vandergaag
Email: Ron_Vandergaag@bcit.ca
Location: SE-8 Rm 127

Course description

This course investigates the different types of DCS and microprocessor controllers such as generic computer control systems. A Foxboro I/A DCS system is investigated and programmed. Students will also gain experience configuring and troubleshooting a PLC using FBD programming. A PC-based HMI (Human Machine Interface) system is also included in this course. Prerequisite: Successful completion of term 3 courses

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Distributed Control Systems

- Explain a typical Foxboro I/A system configuration.
- Explain the Foxboro LAN communication protocols.
- Explain typical hardware components such as AW, WP, CP and associated FBMs.
- Configure various control loops using software tools to program the various components of the Foxboro I/A systems.
- Use System Manager to troubleshoot Foxboro I/A DCS hardware system.
- Troubleshoot a cascade control system complete with operator interface graphics for Foxboro I/A DCS.
- Configure motor control systems using ladder logic in PLB blocks and their associated FBMs.

Functional Block Diagrams

- Explain programming techniques to conform to the IEC 1131 Standards, including Ladder Diagram (LD), Sequential Function Chart (SFC), Function Block Diagram (FBD), Structured Text (ST) and Instruction List (SL).
- Using the FBD programming technique, configure a cascade control system complete with associated motor control applied to a Modicon Quantum PLC.
- Using the SFC programming technique, configure a batch process control system complete with associated motor control applied to a Modicon Quantum PLC.
- Using HMI software interface, troubleshoot the above PLC control system.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

INST 3128 - Boiler Control and Foundation Fieldbus

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	January 08, 2018
End Date:	April 20, 2018
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	85253

Instructor Details

Name: Ron Vandergaag
Email: Ron_Vandergaag@bcit.ca
Location: SE-8 Rm 127

Course description

Students will learn the fundamentals of industrial boiler control, including burner management safety. Students will gain experience tuning up a real industrial boiler control system with state of the art controls to achieve optimum performance from the boiler. Students will gain hands-on experience with

a Foundation Fieldbus control system applied to a wide variety of control loops on a running boiler.
Prerequisite: Successful completion of term 3 courses

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Combustion Control

- Explain the fundamental elements of combustion, Stoichiometric combustion and the importance of the 3 Ts of combustion (time, temperature and turbulence) applied to combustion control.
- Explain the importance, sources and methods of control of CO/NOX/SOX when applied to industrial combustion control.
- Explain the application and operation of flue gas analyzers such as chemical cell, high temperature chemical cell (zirconium oxide), catalytic combustion, paramagnetic etc. applied to industrial boilers.
- Calibrate a microprocessor based O2/COe analyzer applied to industrial boiler control.

Industrial Boilers

- Explain the difference between the fire tube and a water tube industrial boiler.
- Explain the steam components found in an industrial boiler system such as steam drum, mud drum, risers, down comers, de-aerators and the combustion components such as FD fan, ID fan, burner, nozzles, airflow transmitters, gas flow transmitters, gas flow transducers, oil flow transmitters.
- Explain the operation of a Burner Management System (BMS) to conform to ASME, NFPA and BC Safety Authority, industrial gas standards including the operation of primary interlocks and permissive such as low water cut offs, high fuel pressure cut offs, low fuel pressure cut offs, high furnace pressure cut offs, low furnace pressure cut offs, minimum air flow cut offs, low oxygen cut offs, high CO cut offs.

Industrial Boiler Control Systems

- Explain the operation of a parallel open/closed loop, jackshaft and cross limited including excess oxygen trim combustion control systems.
- Explain the difference between plant master and boiler master controls.
- Explain the application of feedforward control indexing to the plant master pressure controller.

- Explain the operation of steam temperature attemporator and conventional desuperheater control.
- Explain the operation of balanced draft furnace pressure control using FD and ID fans.

Foundation Fieldbus Systems

- Explain the fundamental principles of Foundation Fieldbus control compared to conventional analog or DCS control.
- Troubleshoot control loops in a Foundation Fieldbus control system.
- Tune an industrial boiler control system applying a hybrid DCS and Foundation Fieldbus control system.
- Optimize industrial boiler control systems to achieve maximum efficiency and performance under significant load changes.

Gas Analyzers

- Explain the operation and application of gas analyzers such as NOX, SO2, H2S, CO and LEL (lower explosive limit) analyzers.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

INST 3130 - Viscosity, Consistency, Pulp Blend & Batch Control

School:	School of Energy
Program:	Industrial Instrumentation Process and Control Technician
Course Credits:	6
Minimum Passing Grade:	70%
Start Date:	January 08, 2018
End Date:	April 20, 2018
Total Hours:	90
Total Weeks:	15
Hours/Weeks:	6
Delivery Type:	Lecture/Lab
CRN:	85254

Instructor Details

Name: Ron Vandergaag
Email: Ron_Vandergaag@bcit.ca
Location: SE-8 Rm 127

Course description

The accurate measurement of viscosity and consistency is integral to the efficient operation of many industrial processes. In this course, students will learn about different types of viscosity and consistency measurements. Students will calibrate consistency transmitters in the pulp blending lab.
Prerequisite: Successful completion of term 3 courses

Course learning outcomes / competencies

Upon successful completion, the student will be able to:

Safety

- Demonstrate safe working practices in the lab.

Viscosity and Consistency

- Explain the principles of consistency and viscosity devices.
- Calibrate consistency and viscosity transmitters.
- Install consistency and viscosity transmitters.
- Service previously installed consistency and viscosity transmitters.

Pulp Blend

- Start up, operate and shut down a pulp blending process using the DeltaV DCS control system.
- Calculate the consistency of pulp samples.

Batch Control

- Complete a Batch Process (Kraft Digester) using the Foxboro I/A control systems.

Evaluation criteria

Criteria	%	Comments
Weekly Tests	30	• Passing grade is 70% • Students will work on individual projects in labs • There will be no make up tests, exams or quizzes. Exceptions may be made for documented medical reasons or extenuating circumstances. In such a case, it is the responsibility of the student to inform the instructor immediately • Attendance will be taken at the beginning of each session. Students not present at that time will be recorded as absent. If a student is absent for more than 3 days, even with a medical, they may be dismissed from the program • Participation is based on initiative, attitude, workmanship and effective communication in classes and labs
Quizzes	5	
Labs	25	
Participation	10	
Final Exam	30	
TOTAL	100%	

Learning resources

Provided by instructor

BCIT policy

The following statements are in accordance with the BCIT Policies 5101, 5102, 5104, and 7507, and their accompanying procedures. To review these policies and procedures please click on the links below.

Attendance/Illness:
In case of illness or other unavoidable cause of absence, the student must communicate as soon as possible with his/her instructor or Program Head or Chief Instructor, indicating the reason for the absence. Students who are seeking accommodation for a medical absence must have a BCIT approved medical certificate submitted to the department, substantiating the reason for absence. For other absences, the student should be prepared to provide appropriate supporting documentation.

Unapproved absence in excess of the prescribed regulations within this outline may result in failure or forced withdrawal from the course or program. Please see Policy 5101 - Student Regulations, and accompanying procedures.

Academic Integrity:
Violation of academic integrity, including plagiarism, dishonesty in assignments, examinations, or other academic performances are prohibited and will be handled in accordance with Policy 5104 - Academic Integrity and Appeals, and accompanying procedures.

Accommodation:
Any student who may require accommodation from BCIT because of a physical or mental disability should refer to BCIT's Policy on Accommodation for Students with Disabilities (Policy #4501), and contact BCIT's Disability Resource Centre (SW1 2360, 604-451-6963) at the earliest possible time. Requests for accommodation must be made to the Disability Resource Centre, and should not be made to a course instructor or Program area.

Any student who needs special assistance in the event of a medical emergency or building evacuation (either because of a disability or for any other reason) should promptly inform their course instructor(s) and the Disability Resource Centre of their personal circumstances.

Human Rights, Harassment and Discrimination:
The BCIT community is made up of individuals from every ability, background, experience and identity, each contributing uniquely to the richness and diversity of the BCIT community as a whole. In recognition of this, and the intrinsic value of our diversity, BCIT seeks to foster a climate of collaboration, understanding and mutual respect between all members of the community and ensure an inclusive accessible working and learning environment where everyone can succeed.

Campus Mediation Services is a supportive resource for both students and employees of BCIT, to foster a respectful learning and working environment. Any student who feels that they are experiencing discrimination or harassment (personal or human rights-related) can confidentially access this resource for advice and support. Please see Policy 7507 – Harassment and Discrimination and accompanying procedure.

Students should make themselves aware of additional Education, Administration, Safety and other BCIT policies listed at http://www.bcit.ca/about/administration/policies.shtml

Policy for School of Energy

Attempts:Students must successfully complete a course within a maximum of three (3) attempts at the course. Students with two attempts in a single course will be allowed to repeat the course only upon special written permission from the Associate Dean. Students who have not successfully completed a course within three attempts will not be eligible to graduate from their respective program.

Approved

I verify that the content of this course outline is current.
Ron Vandergaag, Instructor
January 05, 2018

I verify that this course outline has been reviewed.
James Armstrong, Chief Instructor
January 05, 2018

I verify that this course outline has been reviewed and complies with BCIT policy.
Brian Buckley, Associate Dean
January 08, 2018

Note: Should changes be required to the content of this course outline, students will be given reasonable notice.